

How do we reliably measure misinformation online?

Luehring, J., Lazzaroni, R.M., Lasser, J.

September 2023

What is being described?

The spread of misinformation has been studied extensively by scholars from various scientific disciplines, therefore, relying on diverse conceptualizations. Definitions of misinformation seem to largely depend on the respective methodological choices, resulting in imprecise or inconclusive findings (Altay et al., 2023). To study the diffusion or discussion of misinformation on social media, computational researchers commonly use stories that were factchecked by experts (Vosoughi et al., 2018), or follow blacklisted domains (González-Bailón et al., 2023; Guess et al., 2019; Grinberg et al., 2019; Robertson et al., 2023). The choice between story and source-based approaches hinges on the primary research objective. When operating at the level of the story, we capture misinformation with high precision, but the selection of stories can be narrow and is likely biased. Conversely, with a source-based strategy, we can cover a wider variety of sources and stories, and hence, the data may be less biased. However, this approach runs the risk of overgeneralizing across stories and source ratings may be especially sensitive to time and context. Therefore, we review different approaches to measuring misinformation online in terms of reliability, validity and feasibility and then focus on one perspective in particular that has recently gained popularity: domain-based approaches.

More precisely, this proposal outlines research built on the question to what extent source-based approaches can reliably measure misinformation online (over time and across contexts).

How is the sample constructed?

In the United States, one of the most prominent lists used for assessing misinformation is curated by the organization NewsGuard. NewsGuard’s analysts assign ratings based on nine distinct criteria and regularly update these assessments. The list has been increasingly used and research based in top-tier journals (Robertson et al., 2023). However, there are several challenges associated with this list: First, the domain ratings were initially created as a tool for

brand security rather than a research instrument, which is why the list is proprietary and access as well as publication of the data is restricted. Second, the coverage of the list has so far only been validated for research purposed for the U.S. context (Lin et al., 2023), but it is crucial to assess the usefulness of this list for research beyond the English-speaking world. Third, the list is updated in unknown intervals, which is why it is important to understand the update process and its implications for research. Fourth, the list is not exhaustive, which is why it is important to understand the selection process and its implications for research. Fifth, the list is not free of errors and missing data points, which is why it is important to understand their implications for research.

Consequently, we propose to provide a comprehensive description of this database over time (monthly granularity) and across countries provided by NewsGuard. We believe that is essential to address the following aspects of the database:

- a) the distribution, construction, weighting and update process of the ratings,
- b) the selection and removal of sources, and
- c) the completeness and reliability of other information provided in the database, e.g., political orientation.

In addition, we reproduce published research relying on the NewsGuard list with different versions of the list to assess the degree to which results depend on a specific version of the list.

How does it pertain to digital media?

The ramifications of misinformation are multifaceted and may range from political polarization (Guess et al., 2023) to vaccine hesitancy (Brennen et al., 2023). Consequently, it is imperative to understand the spread of misinformation and its interactions with digital media and society. In order to do so, computational social scientists need to have access to reliable and valid data. By reviewing strategies to research misinformation online, and assessing the reliability of source-based approaches in particular, we outline potential drawbacks and pitfalls for digital media research and offer recommendations for the application of the NewsGuard database in different contexts.